1. Evaluate the following line integrals with respect to arc length: (1) $\int_L \sqrt{y}\,\mathrm{d}s$, where L is the arc of the parabola $y = x^2$ from the origin $(0,0)$ to the point $(1,1)$

(2) $\oint_\Gamma (x^2+y^2+z^2)\,\mathrm{d}s$, where Γ is the arc of the helix $x = \cos t, y = \sin t, z = t$ corresponding to $t = 0$ to $t = 2\pi$

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2. Evaluate the following line integrals with respect to coordinates: (1) $\int_L x\mathrm{d}y - y\mathrm{d}x$, where L is the arc of the cycloid $x = t - \sin t,\ y = 1 - \cos t$ from the point $(0,0)$ to the point $(2\pi, 0)$

(2) $\oint_L \frac{(x+y)\mathrm{d}x-(x-y)\mathrm{d}y}{x^2+y^2}$, where L is the circle $x^2+y^2 = a^2\ (a>0)$ traversed counterclockwise

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3. Evaluate the following curve integrals: (1) $\oint_L (2xy - x^2)dx + (x+y^2)dy$, where L is the positively oriented boundary curve of the closed region bounded by the parabolas $y = x^2$ and $y^2 = x$

(2) $\int_L (e^x \sin y - 2y)dx + (e^x \cos y - 2)dy$, where L is the upper semicircle $(x-a)^2 + y^2 = a^2\ (y \geq 0)$ traversed clockwise

4. Using line integrals, find the area of the region bounded by the circle $x^2 + y^2 = 2y$.

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5. Evaluate the following surface integrals with respect to area: (1) $\iint_\Sigma \sqrt{1+4z}\,\mathrm{d}S$, where Σ is the part of the paraboloid $z = x^2 + y^2$ with $z \leq 1$

(2) $\iint_\Sigma \frac{1}{x^2+y^2+z^2}\,\mathrm{d}S$, where Σ is the part of the cylinder $x^2+y^2 = R^2\ (R>0)$ between the planes $z = 0$ and $z = H\ (H>0)$

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6. Evaluate the following surface integrals with respect to coordinates: (1) $\iint_\Sigma z\,\mathrm{d}x\mathrm{d}y$, where Σ is the lower side of the cone $z = \sqrt{x^2+y^2}\ (0 \leq z \leq 1)$

(2) $\iint_\Sigma (x+1)\mathrm{d}y\mathrm{d}z + y\mathrm{d}z\mathrm{d}x + \mathrm{d}x\mathrm{d}y$, where Σ is the entire boundary surface of the spatial region bounded by the planes $x = 0, y = 0, z = 0, x + y + z = 1$, oriented outward

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7. Evaluate the following surface integrals using Gauss's theorem: (1) $\oiint_\Sigma 2xz\mathrm{d}y\mathrm{d}z + yz\mathrm{d}z\mathrm{d}x - z^2\mathrm{d}x\mathrm{d}y$, where Σ is the outer side of the boundary surface of the region bounded by the cone $z = \sqrt{x^2+y^2}$ and the hemisphere $z = \sqrt{2-x^2-y^2}$

(2) $\iint_\Sigma x\mathrm{d}y\mathrm{d}z + y\mathrm{d}z\mathrm{d}x + z\mathrm{d}x\mathrm{d}y$, where Σ is the upper side of the upper hemisphere $z = \sqrt{1-x^2-y^2}$